

Changyu (Thomason) Zhao

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EDUCATION

Stanford University <i>M.S. in Computer Science (Computer and Network Security)</i>	Sep. 2025 – Jun. 2027 (expected) Stanford, California
University of Wisconsin-Madison <i>B.S. in Computer Science (Honors) and Mathematics</i>	Sep. 2021 – May 2025 Madison, Wisconsin
GPA: 3.95/4.0 , David Dewitt Undergraduate Scholarship, Dean's List (all semesters)	

EXPERIENCES

UW-Madison Cybersecurity Operation Center <i>Cybersecurity Student Intern Analyst</i>	Feb. 2023 – May 2025 Madison, Wisconsin
- Investigated 1000+ security incidents (malware, phishing, intrusions) using Elastic SIEM. Developed threat hunting queries and detection rules that reduced MTTR by 40% and repeat infections by 25%. - Conducted monthly IAM attestations for 500+ accounts using automated scripts, remediating 3% toxic access in 24h.	
UW-Madison MadS&P Lab <i>Research Assistant, Advisor: Prof. Patrick McDaniel</i>	Sep. 2024 – Aug. 2025 Madison, Wisconsin
- Conducted comprehensive evaluation of LibIHT against Intel Pin and DynamoRIO on real-world workloads, demonstrating 150x speedup, 99%+ code coverage, and complete resistance to anti-debugging techniques employed by malware. - Established reproducible benchmarking infrastructure with 21 coreutils binaries and 10 malware specimens, automating performance metrics collection and visualization in Python. Paper accepted by ACM CCS 2025 SURE workshop.	
ASU SEFCOM Lab <i>Research Assistant, Advisor: Prof. Yan Shoshitaishvili and Ruoyu Wang</i>	May. 2024 – Sep. 2024 Tempe, Arizona
- Co-designed Oxidizer, the first Rust-focused decompiler built atop angr, implementing control-flow recovery, function prototype inference, and macro inference to address the unique challenges of analyzing memory-safe compiled binaries. - Developed static type recovery engine to extract Rust-specific type metadata from compiled binaries and integrates findings into decompilation, improving accuracy of data structure and function signature reconstruction for security analysis.	
Tencent Xuanwu Lab <i>Security Researcher Intern</i>	Jul. 2023 – Aug. 2023 Beijing, China
- Architected LibIHT, a cross-platform dynamic binary tracing framework in C/C++/Python that leverages Intel LBR/BTS hardware features for control-flow tracing, achieving 150x speedup over Intel Pin on large-scale program analysis. - Designed unified kernel driver architecture for Linux and Windows with abstraction layers for OS and hardware, reducing code duplication by 40% while standardizing trace session management and improving cross-platform maintainability. - Implemented user-space API library and debugger plugin system for GDB/WinDbg integration, enabling security analysts to visualize execution traces, identify anomalous control-flow patterns, and accelerate vulnerability discovery workflows.	
ASU SEFCOM Lab <i>Research Assistant, Advisor: Prof. Yan Shoshitaishvili</i>	May. 2023 – Jul. 2023 Tempe, Arizona
- Developed LibHopper, an automated exploitation primitive discovery tool combining debugger instrumentation with symbolic execution to identify exploitable memory corruption opportunities in library state structures. - Constructed exploitation primitives dataset spanning critical C libraries (OpenSSL, OpenSSH, libcurl, libpng), providing structured risk assessments to inform security research in automated exploit generation and vulnerability scanning tooling.	

PROJECTS

Router automated vulnerability discovery <i>IoT, Reverse Engineering, Web Security</i>	May 2022 – Aug. 2022
- Built an IDA plugin for static IoT firmware analysis, uncovering 7 unfixed CVEs in the Tenda AC18 router. - Performed dynamic analysis with QEMU rehosting and GDB to develop PoC exploits for identified vulnerabilities.	

PRESENTATIONS & PUBLICATIONS

Yibo Liu, . . . , Changyu Zhao , . . . “Oxidizer: Toward Concise and High-fidelity Rust Decompilation”. <i>2026 IEEE Symposium on Security and Privacy (IEEE S&P)</i> . May 2026.
Changyu Zhao , . . . “LibIHT: A Hardware-Based Approach to Efficient and Evasion-Resilient Dynamic Binary Analysis”. <i>Proceedings of the 1st Workshop on Software Understanding and Reverse Engineering (SURE 2025)</i> . ACM, October 2025.
Black Hat Arsenal, <i>LIBIHT: A Cross-Platform Library for Accessing Intel Hardware Trace Features</i> , Las Vegas, NV, Aug. 2024

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team <i>Team Member</i>	Sep. 2021 – Present International
- Entered DEFCON CTF world final from 2022 to 2025, ranked 12th, 7th, 4th, and 5th respectively. - Competed in 120+ CTF competitions, expert in kernel and user-space exploitation and reverse engineering.	

SKILLS

Languages: C/C++, Python, Rust, Go, Java, x86/x64, ARM, MIPS Assembly
Systems & Reverse Engineering: QEMU, GDB, WinDbg, IDA Pro, Ghidra
Security Analysis: Angr, AFL++, Frida, Joern, Wireshark, Metasploit, Kali